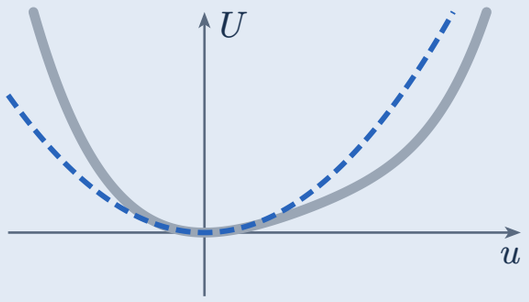
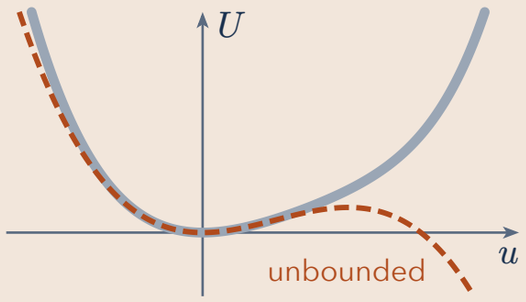
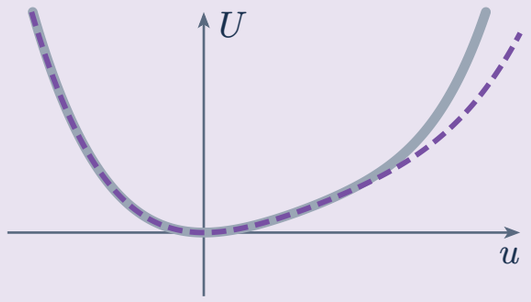
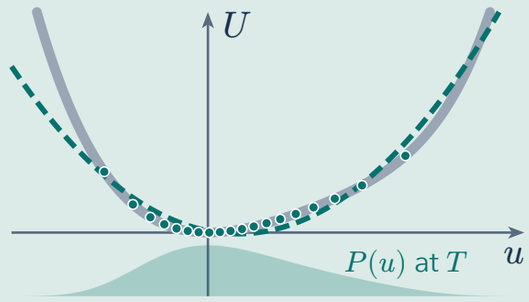
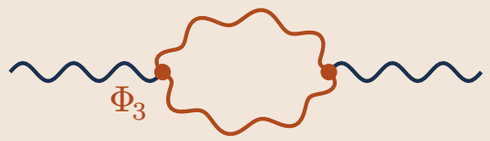
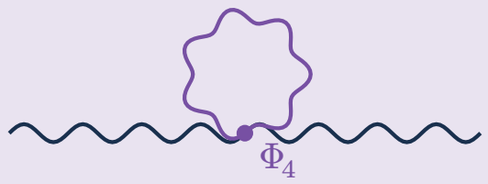
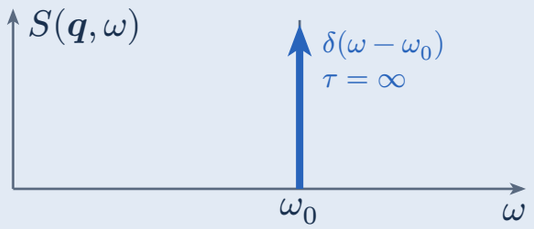
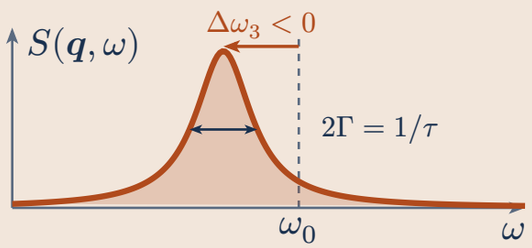
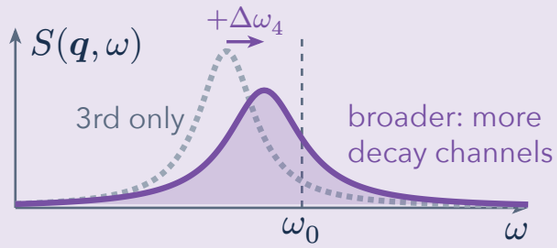
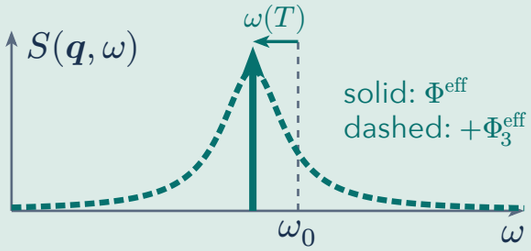
Perturbative: Taylor-expand U in displacements u about the static $T = 0$ K lattice, one term per column

Effective: fit to MD at temperature T

	Harmonic	+ 3rd order (cubic)	+ 4th order (quartic)	TDEP from MD
	$U = U_0 + \frac{1}{2!}\sum_{ij}\Phi_{ij}u_iu_j$	$+\frac{1}{3!}\sum_{ijk}\Phi_{ijk}u_iu_ju_k$	$+\frac{1}{4!}\sum_{ijkl}\Phi_{ijkl}u_iu_ju_ku_l + \ldots$	$U \approx U_0(T) + \frac{1}{2!}\sum_{ij}\Phi^{\text{eff}}_{ij}(T)u_iu_j$
Potential	 gray: true $U(u)$ dashed: $\frac{1}{2}\Phi_2u^2$	 dashed: captures the skew	 dashed: bounded, good for small u	 dots: MD snapshots dashed: fit at T
Ingredients	 Independent normal modes from $\det(D(\boldsymbol{q}) - \omega^2 I) = 0$: free phonons that never scatter	 Bubble: a phonon decays into two (and back), giving lifetime τ and a downshift	 Loop: shift $\propto \Phi_4\langle u^2\rangle \propto T$, same order as the bubble; plus 4-phonon scattering	 Fitting $\min\sum_t \lvert\boldsymbol{F}_t^{\text{MD}} - \boldsymbol{F}_t^{\text{model}}\rvert^2$ folds all orders into $\Phi^{\text{eff}}(T)$
Line shape				
Trade-offs	**Input** Φ_2 from finite displacements or DFPT on the static lattice (e.g. phonopy). **Captures** dispersions, DOS, zero-point energy, C_V and vibrational free energy; the quasi-harmonic approximation adds volume dependence. **Misses** ω is independent of T, lifetimes are infinite ($\kappa_L \to \infty$), and phases stabilized only by temperature (bcc Zr, cubic $SrTiO_3$) show imaginary modes.	**Input** Φ_3 from $\propto N^2$ displaced supercells (phono3py, ShengBTE). **Captures** three-phonon decay and Umklapp scattering: lifetimes τ, linewidths $\Gamma \propto T$ at high T, lattice thermal conductivity κ_L; the bubble softens ω. **Misses** the cubic potential alone is unbounded, so it only works as a perturbation around $T = 0$ K modes, and fails where those are imaginary.	**Input** Φ_4 needs $\propto N^3$ displacements, often fit by compressive sensing (hiPhive, ALAMODE). **Captures** loop shift of $\omega(T)$, the same order in $\langle u^2\rangle$ as the bubble; four-phonon scattering lowers κ_L (e.g. BAs); self-consistent phonons can stabilize imaginary modes. **Misses** still a truncated expansion: strongly anharmonic or diffusive systems need higher orders, and cost grows steeply.	**Input** MD at temperature T (ab initio or ML potential), then a linear least-squares fit of Φ^{eff} to the sampled forces. **Captures** all orders at the thermal amplitudes: renormalized $\omega(T)$, high-T phases stabilized, average positions from MD; fitting Φ_3^{eff}, Φ_4^{eff} to the same data gives lifetimes. **Misses** needs MD at every T; classical nuclei unless sampled quantum mechanically (PIMD, stochastic sTDEP); Φ^{eff} is a fit, not a Taylor coefficient.